WEEK 6 – POWER BI HEALTHCARE DATA ANALYSIS

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INTRODUCTION:

This project was continued using Power BI to perform analysis and visualization on the prepared healthcare dataset. After completing the data cleaning, transformation, dimension tables, Billing table, and data model, DAX measures were created to calculate important healthcare and billing values.

The calculated measures were then used to construct KPI cards, matrix visuals, donut charts, bar charts, line charts, and clustered column charts. Slicers were also added to provide dynamic filtering and cross-filtering of the dashboard.

OBJECTIVES:

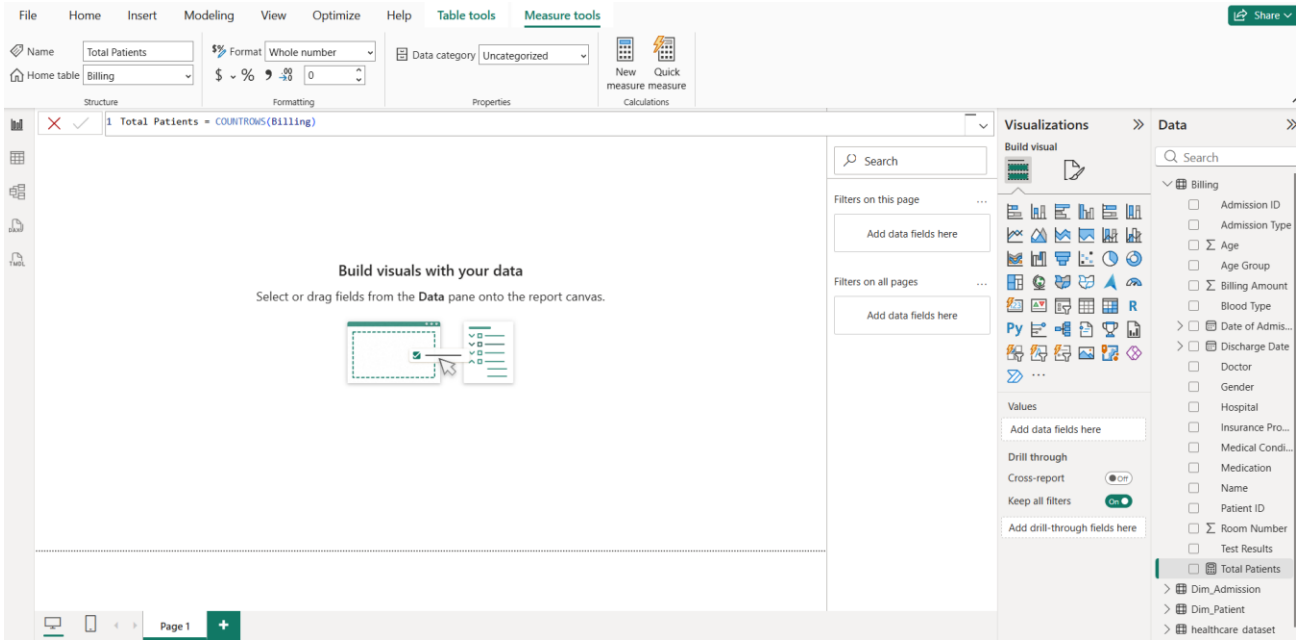
The main objective of the project was to create meaningful healthcare measures and visualizations using the prepared data model. DAX measures were created for total patients, total billing amount, average billing amount, average length of stay, and readmission rate.

These measures were then used to construct dashboard elements for demographic analysis, medical-condition analysis, admission trends, hospital-related analysis, and insurance-provider analysis. Slicers were finally added to make the dashboard interactive and allow users to filter the visualizations dynamically.

DAX MEASURES:

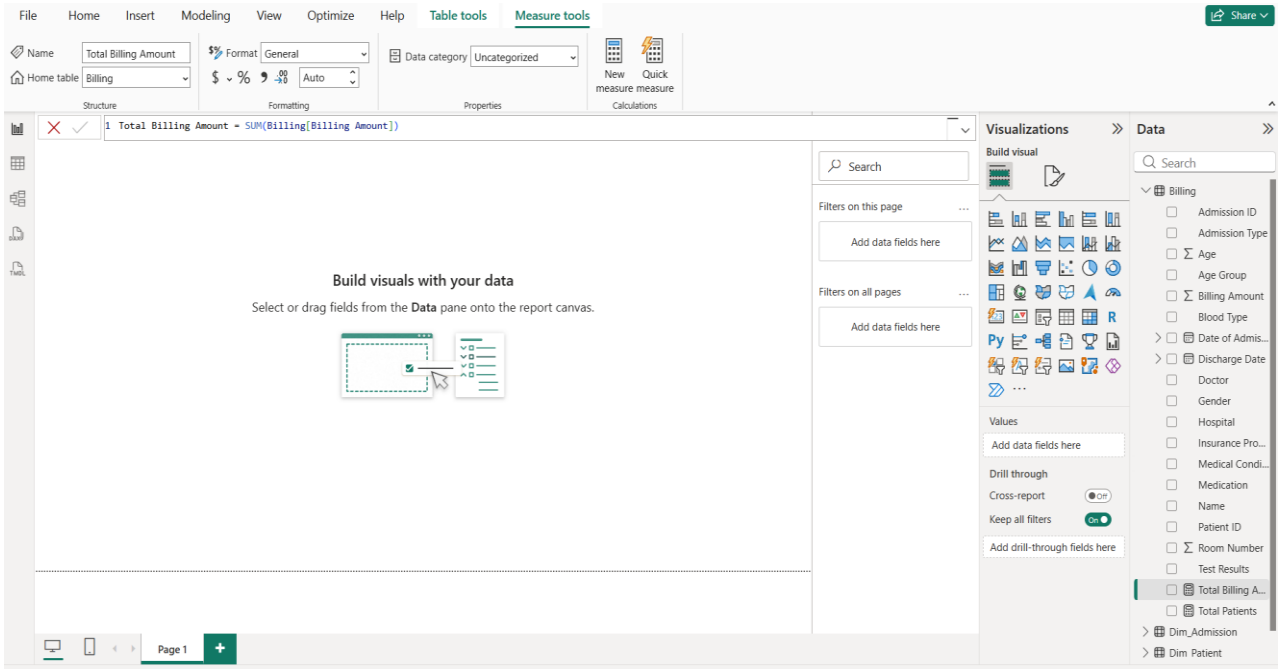
CREATING TOTAL PATIENTS MEASURE:

A new measure was created in the Billing table to calculate the total number of patient records. The COUNTROWS function was used to count the rows available in the Billing table. The measure was named Total Patients and was then used in the KPI card and other dashboard visuals.



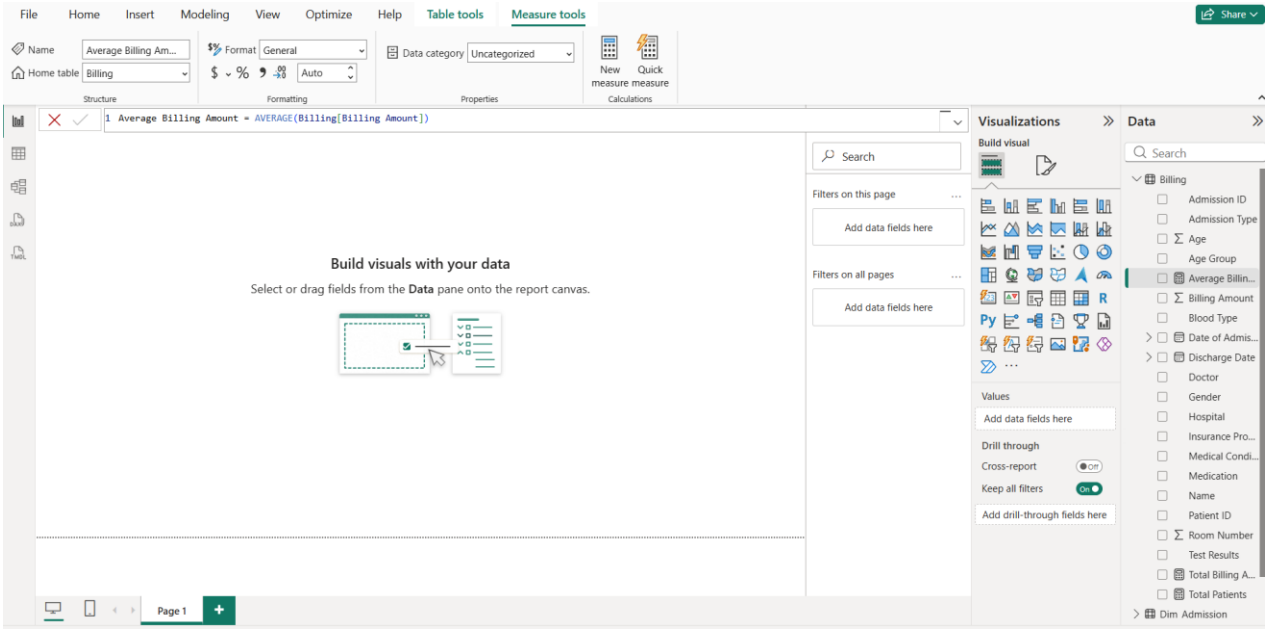
CREATING TOTAL BILLING AMOUNT MEASURE:

A new measure was created to calculate the total billing amount from the healthcare data. The SUM function was applied to the Billing Amount column in the Billing table. The measure was named Total Billing Amount and was used in the KPI card and billing-related visualizations.



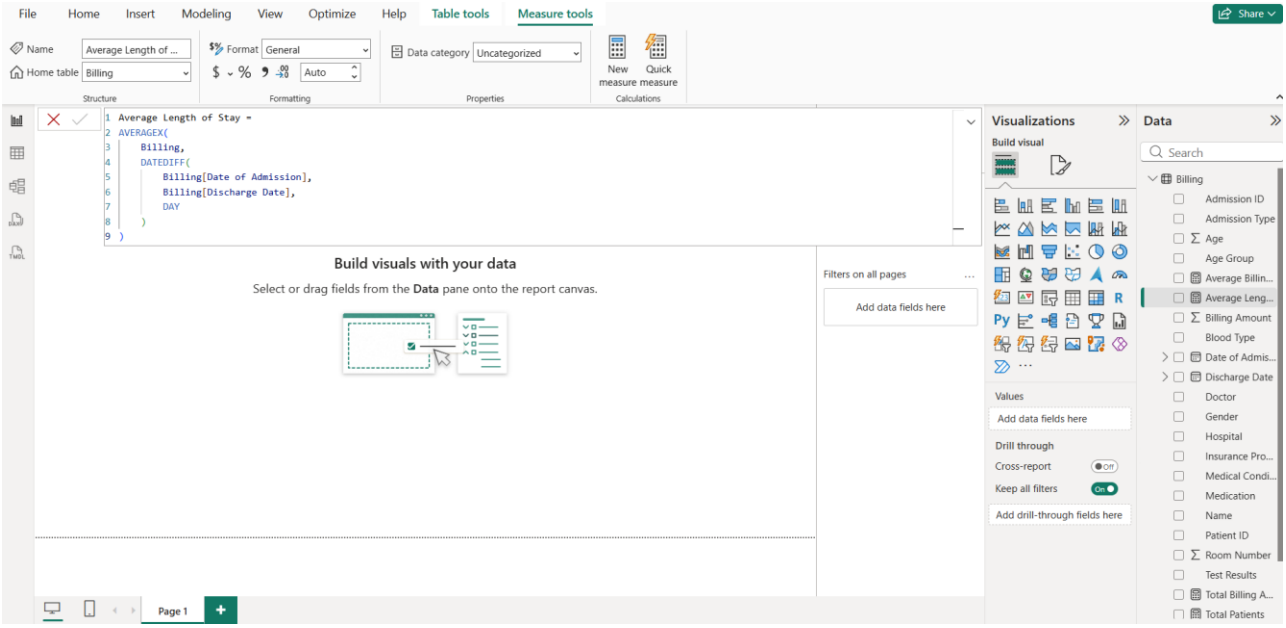
CREATING AVERAGE BILLING AMOUNT MEASURE:

A new measure was created to calculate the average number of days patients stayed in the hospital. The DATEDIFF function was used to find the difference between the Date of Admission and Discharge Date, and AVERAGEX was used to calculate the average. The measure was named Average Length of Stay and was used in the KPI card.



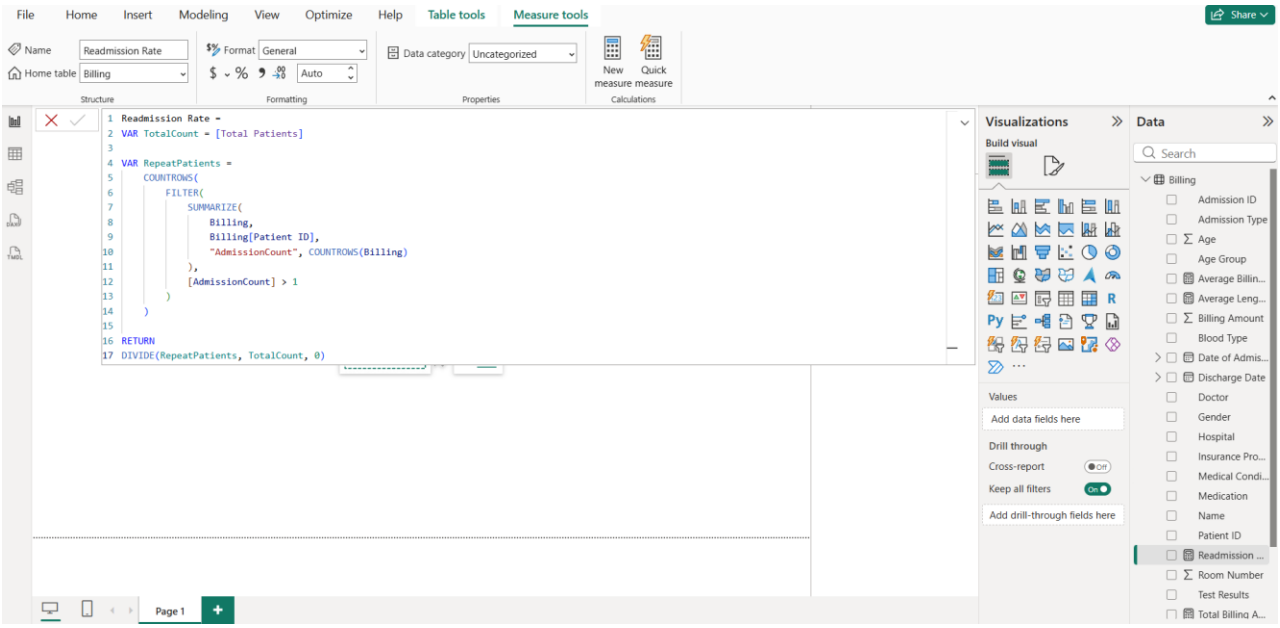
CREATING AVERAGE LENGTH OF STAY MEASURE:

A new measure was created to calculate the average number of days patients stayed in the hospital. The DATEDIFF function was used to find the difference between the Date of Admission and Discharge Date was used to calculate the average. The measure was named Average Length of Stay and was used in the KPI card.



CREATING READMISSION RATE MEASURE:

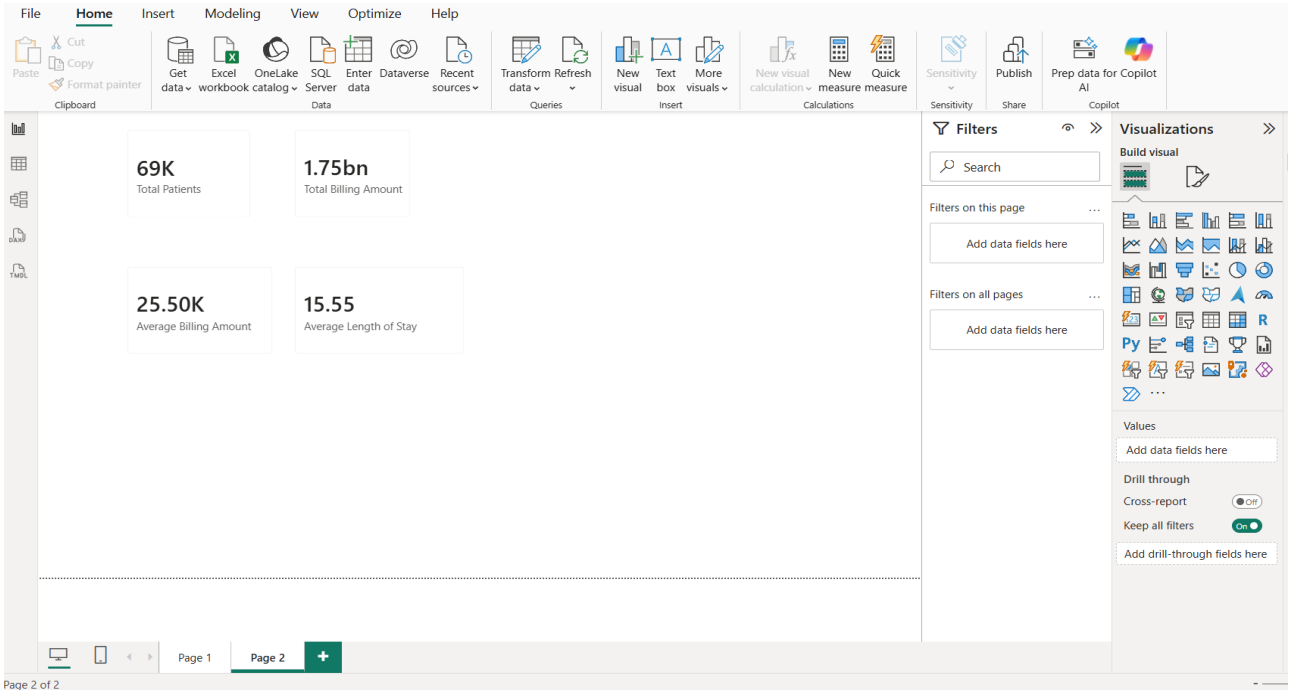
A new measure was created to calculate the readmission rate of patients. The measure identified patients with more than one admission and compared the number of repeat patients with the total patient count. The DIVIDE function was used to calculate the rate, and the measure was named Readmission Rate for further analysis.



CONSTRUCTING VISUAL LAYOUT AND DASHBOARD ELEMENTS:

CREATING TOP KPI SUMMARY CARDS:

After creating the DAX measures, four KPI summary cards were created at the top of the Power BI report. A Card visual was selected from the Visualizations pane and the Total Patients measure was added to the first card. Similar Card visuals were created for Total Billing Amount, Average Billing Amount, and Average Length of Stay.



CREATING MATRIX VISUAL:

A Matrix visual was created to analyze patient distribution based on age group, gender, and medical condition. Age Group and Gender were added to Rows, while Medical Condition was added to Columns. Total Patients and Average Billing Amount were added to Values to compare patient counts and billing amounts across different groups.

CREATING THE DONUT CHART:

A Donut Chart was created to show the distribution of patients across different age groups. The Age Group field was added to the Legend and Total Patients was added to Values. The chart visually represented the number of patients belonging to each age group.

CREATING THE STACKED BAR CHART:

A Stacked Bar Chart was created to analyze patient numbers for different medical conditions. Medical Condition was placed on the Y-axis and Total Patients on the X-axis. Admission Type was added to separate the patient counts according to different types of admission.

CREATING THE LINE CHART:

A Line Chart was created to analyze admission trends over time. Date of Admission was added to the X-axis using the Year and Month hierarchy. Total Patients and Total Billing Amount were added to the Y-axis to display changes in admissions and billing over time.

CREATING THE CLUSTERED COLUMN CHART:

A Clustered Column Chart was created to compare billing amounts across different insurance providers. Insurance Provider was placed on the X-axis and Total Billing Amount was placed on the Y-axis. The chart displayed separate columns for each insurance provider for easy comparison.

CREATING THE INSURANCE PROVIDER SLICER:

A Slicer visual was created to filter the dashboard based on insurance providers. The Insurance Provider field was added to the slicer and arranged with the other dashboard filters. Selecting an insurance provider dynamically updates the related visuals in the report.

CREATING THE ADMISSION TYPE SLICER:

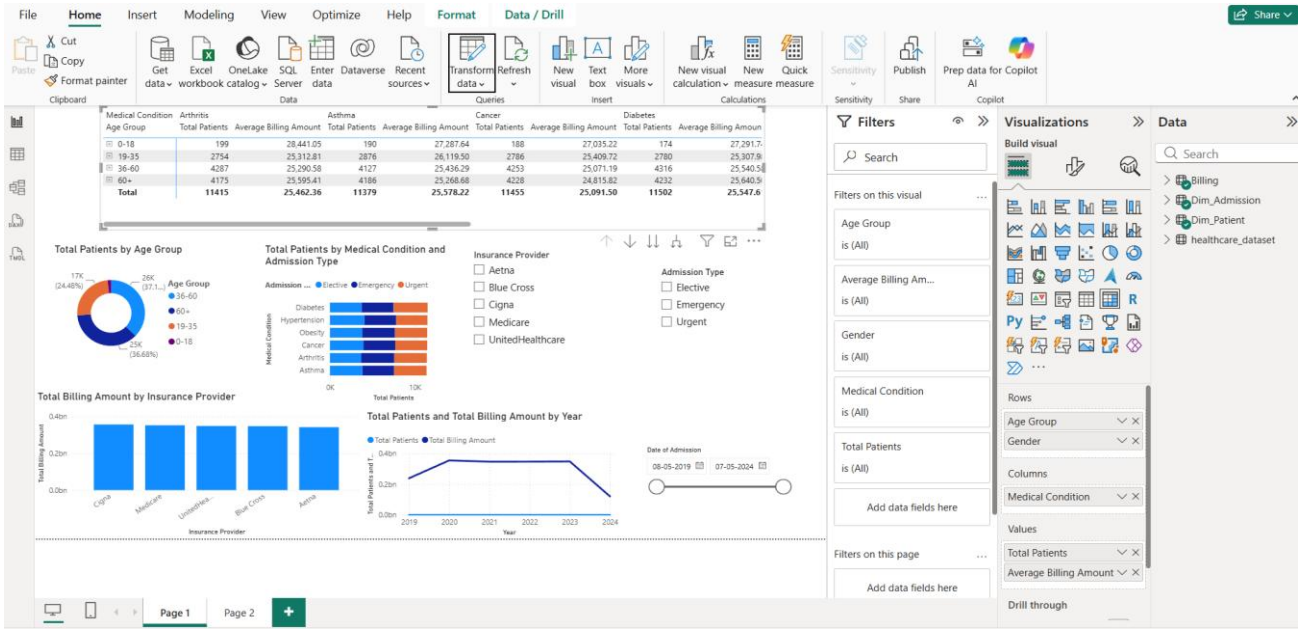
A second Slicer visual was created to filter the dashboard according to admission type. The Admission Type field was added to the slicer and positioned in the dashboard. This allows the required admission type to be selected and the related visual information to be displayed.

CREATING THE DATE RANGE SLICER:

A third Slicer visual was created using the Date of Admission field. The slicer was configured as a date-range filter to allow a specific period to be selected. The dashboard visuals are then updated according to the selected admission dates.

FINAL DASHBOARD:

After creating all the measures and visuals, they were arranged into a structured dashboard layout. The KPI cards were placed at the top, followed by the Matrix, Donut Chart, Stacked Bar Chart, Line Chart, and Clustered Column Chart. The three slicers were also positioned on the dashboard to provide interactive filtering across the visuals.



CONCLUSION:

The project successfully demonstrated the use of Power BI for healthcare data analysis and visualization. DAX measures were created and used with different visualizations to present patient, billing, admission, and demographic information. Interactive slicers were also added, resulting in a structured and user-friendly healthcare analysis dashboard.